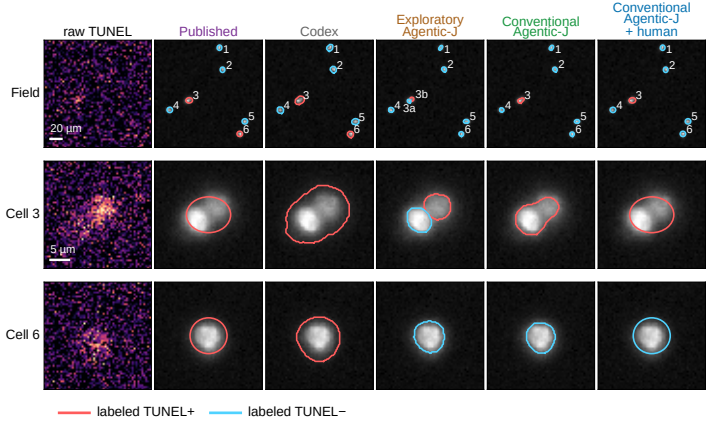
a

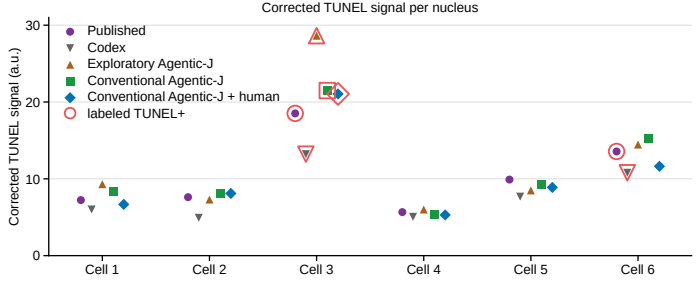
	Registration (live → fixed)	Host / nucleus segmentation	TUNEL call	Parasite localisation	Measurement & statistics	Time T1 / T2	Task 1 TUNEL classification (6 nuclei, fixed frame)	Task 2 proliferation, Fig. 4C (15 parasites, last live frame)
Published	by hand	hand-drawn ROIs	by inspection	15 ROIs + 3 backgrounds each, by hand	ROI Manager → Excel; Mann–Whitney	not recorded	2/6 nuclei TUNEL+	8 TUNEL+ vs 7 TUNEL−, p = 0.0012
Codex CLI	SIFT + RANSAC, ECC refined (1.6 px)	T1 DAPI threshold; T2 Cellpose cyto3	largest-gap cutoff	red-channel peaks + water-shed: 10 kept	annulus background; no test	18 min (both)	✓ 6/6 found, 6/6 calls, r = 0.98	✓ 10/15 found, 10/10 hosts right; p = 0.017
Exploratory Agentic-J	TurboReg rigid on brightfield	T1 StarDist; T2 Cellpose cyto3 (7 hosts)	largest-gap cutoff	C2 + C3 spot evidence: 22 kept	annulus background; no test	22 / 69 min	✓ 6/6 (+1 split), 5/6 calls, r = 0.95	✗ 0/15 found (+22), no host
Conventional Agentic-J	TurboReg / StackReg rigid	T1 DoG + water-shed; T2 Cellpose cyto3 (6 hosts)	2-component GMM	micro-sam vit_b_lm: 105 → 5 kept	annulus background; no test	55 / 85 min	✓ 6/6 found, 5/6 calls, r = 0.99	✗ 3/15 found (+2), no TUNEL+
Conventional Agentic-J + human	lab ROIs given, origin translated	lab ROIs (given)	largest-gap cutoff	lab ROIs paired 1:1 with backgrounds	native 16-bit; Mann–Whitney	32 min / n.r.	✓ 5/6 calls agree, r = 0.97	✓ 15/15; 14/15 within 0.1; p = 0.0012

Manual · Human feedback · Automatic (Agentic-J) · Automatic (other agent)

b

Task 1: the same six nuclei, segmented and called by each pipeline





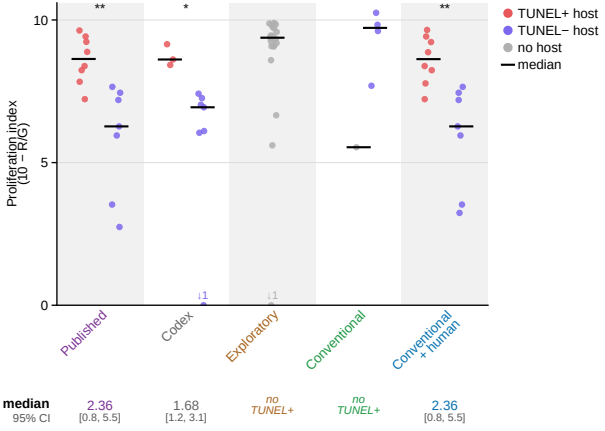
c

Agreement with the published analysis

		Codex	Exploratory Agentic-J	Conventional Agentic-J	Conventional Agentic-J + human
Task 1	TUNEL calls agree	6 / 6	5 / 6	5 / 6	5 / 6
Task 2	Lab parasites found	10 / 15	0 / 15	3 / 15	15 / 15
	Host group correct	10 / 10	–	2 / 3	15 / 15
	Index within 0.1 of lab	4 / 10	–	1 / 3	14 / 15
	False detections	0	22	2	0

d

Task 2: proliferation index distributions



	Published	Codex	Exploratory	Conventional	Conventional + human
Δ median (95% CI)	2.36 [0.8, 5.5]	1.68 [1.2, 3.1]	no TUNEL+	no TUNEL+	2.36 [0.8, 5.5]